

SAI PRANAV REDDY

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[LinkedIn](#) | [GitHub](#) | [LeetCode](#) | [Codeforces](#) | [CodeChef](#)

Education

Gayatri Vidya Parishad College of Engineering

B.Tech in Information Technology | GPA: 8.51/10

2024 - 2028

Achievers Junior College

Intermediate in MPC | Percentage: 80.6%

2021 - 2023

Experience

Research Intern, National Institute of Technology Raipur

May 2026 - July 2026

- Built a hierarchical graph-based reinforcement learning framework for cooperative UAV/UGV task allocation and path planning in dynamic, constraint-heavy environments.
- Designed adaptive graph-update algorithms using obstacle, terrain, communication, and battery-state feedback to improve routing quality and system resilience.
- Developed a benchmarking pipeline with dynamic obstacles, mission priorities, communication events, and battery constraints for reproducible performance evaluation.
- Improved multi-objective mission performance by 7.2% and reduced collision risk by 58.1% across 9 benchmark scenarios.
- Implemented simulation systems, graph algorithms, path-planning logic, data analysis scripts, and performance evaluation tooling in Python.

Projects

OptiRake DSS - AI-Based Rake Planning Optimizer

[Deployment](#)

- Built a decision-support system for SAIL Bokaro Steel Plant to optimize rail-rake formation logistics across 100+ customer orders and operational constraints.
- Designed optimization logic to process 6 CSV inputs covering orders, rakes, stockyards, product-wagon compatibility, loading points, and route costs.
- Achieved 80–90% wagon utilization, reduced rake requirement by 33%, improved throughput by 130%, and reduced cost per order by 20%.
- Reduced manual planning time from 2–3 hours to nearly 2 seconds through automated assignment, explainable KPI cards, and rake-level reasoning.
- Built full-stack application using React 18, TypeScript, Tailwind CSS, Express.js, REST APIs, and CSV parsing.

CodeStruct – Comprehensive DSA Learning & Performance Platform

[Repo](#)

- Developed a full-stack DSA platform integrating live competitive-programming performance from LeetCode, Codeforces, and CodeChef into a unified dashboard.
- Engineered structured DSA pattern roadmaps and language-specific learning paths covering C++, Java, and Python for systematic skill progression.
- Built a responsive Next.js/React application with TypeScript, Tailwind CSS, Framer Motion, and Supabase for interactive learning workflows, data persistence, and performance tracking.
- Implemented and iteratively maintained the platform across 100+ Git commits, integrating modular UI components, backend services, and database-backed application features.

Adaptive Feedback Driven Hierarchical Graph Reinforcement Learning for UAV/UGV Missions

[Repo](#)

- Developed a feedback-driven hierarchical graph reinforcement learning research paper for cooperative task allocation and adaptive path planning.
- Compared static graph, centralized feedback with static topology, and adaptive feedback HGRL baselines using scenario-wise metrics.
- Reduced mean objective score from 79.26 to 73.54 and mean collision risk from 0.0609 to 0.0255 across 9 scenarios.
- Generated research-style outputs including metrics CSVs, aggregate summaries, objective graphs, risk graphs, battery graphs, and adaptive path visualizations.

Skills

Languages: C++, Python, JavaScript, C, HTML, CSS

Computer Science: Data Structures, Algorithms, Object-Oriented Programming, Operating Systems, Database Management Systems, Computer Networks

Software Engineering: Software Design, REST APIs, Debugging, Backend Development, Full-Stack Development, Performance Evaluation, System Design Fundamentals

Frameworks and Libraries: React.js, Next.js, Express.js, Node.js, Tailwind CSS

Databases and Tools: MongoDB Atlas, Supabase, Git, GitHub, VS Code, CSV Processing, API Integration

Achievements and Roles

- **Smart India Hackathon Finalist:** Selected among the **top 1,300 teams from 50,000+ teams nationwide**.
- **Competitive Programming:** **LeetCode Knight** (1871) | **Codeforces Pupil** (1219) | **CodeChef 2*** (1492).
- **LeetCode Weekly Contest 511:** Secured **1208th Rank**, Among more than 40,000 participants
- **AlgoRhythm LeetCode League:** Secured **2nd place** college-wide through consistent Problem of the Day (POTD) participation, strong performance in coding contests over a month, and success in the final technical interview.
- **Pixel Perfect Challenge 2024:** Secured **3rd place** for buiding a prototype website of a car accessories business within a **1-hour frontend challenge** in college.